QUINN YOCKEY

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EDUCATION

Honors BS in Electrical Computer Engineering, Minor in Computer Science — Expected June 2027
Oregon State University — *Corvallis, OR*

- 3.96 GPA
- Presidential Scholar
- Honor Roll & Engineering Dean's List Fall 2023 – Present
- Tau Beta Pi Engineering Honor Society Member

PROFESSIONAL EXPERIENCE

Electrical Engineer Intern – Wisp Project Lead — Jan. 2026 – Present
OPEnS Lab — *Corvallis, OR*

- Assemble electrical stacks, troubleshoot & deploy "Wisp" IoT air quality sensing devices in 44 vineyards.
- Debug hardware performance, power & race conditions, collaborating closely with software team.
- Lead weekly lab trainings teaching integration of sensors into embedded C++ codebase.
- Simplify areas of high software complexity and fight feature creep.
- Harden software and hardware designs for stability in months-long outdoor deployments.

Software Engineer Intern — Jun. – Sep. 2025
Wind River — *San Diego, CA*

- Developed python tool to collect data on internal project status using SOAP API from tracking database.
- Automated progress data aggregation into spreadsheets, saving hours per week for project managers of multi-million-dollar commissions.

Embedded Software & Electrical Technician Intern — Jul. - Sep. 2024
Lime Rock, LLC — *Medford, OR*

- Designed serial communication interface in embedded C for STM32 microcontroller, enabling seamless and abstract integration with measurement sensor.
- Utilized oscilloscope for extensive debugging of hardware and software.
- Applied quaternion mathematics to implement precise stabilization feedback, mitigating gimbal lock risk.
- Increased manufacturing efficiency of military-spec wiring harnesses & soldered circuit board assemblies.
- Collaborated with engineers to write and refine documentation throughout prototype development.

PROJECTS

Oscilloscope — Jan. – Mar. 2026

- Designed and constructed 2-channel digital oscilloscope, collaborating with team of 3 students.
- Developed embedded software in C to interface with ADC at fast 200 kHz resolution.
- Integrated C software modules with efficient control flow.
- Integrated module schematics into PCB layout with attention to analog signal integrity.

Bicycle Speedometer — Oct. 2024 – Jan. 2025

- Designed and implemented bike speedometer for ATmega32U4 microcontroller.
- Iterated over software architecture using embedded C, ultimately reaching interrupt-driven control flow.
- Optimized design for low power consumption to maximize device battery life.
- Developed error-resistant drivers for serial transmission and OLED display.

SKILLS

Languages: C, C++, Python, SystemVerilog, Assembly (AVR & 6502), Java.
Tools: Linux, Git, Bash, Make, GDB, Arduino, Valgrind, Quartus Prime, ModelSim, Vim, STM32 IDE.
Concepts: Embedded systems, Microcontrollers, Sensors, FPGA design, Debugging.
Soft Skills: Communication, Motivation, Datasheet parsing, Documentation, Detail-oriented thinking.